

Sparsh Srivastava

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Summary

AI Developer who has engineered production **AI systems** and **voice AI agents** using **Python**, **LangChain**, and **vector databases**, applying **retrieval-augmented generation** and **large language model** integration across web-connected applications with **PostgreSQL**-backed data design. Benchmarked **1,000+** **AI tools** in a research role and placed **1st** in a **100+ participant** AI hackathon. Backed by **3** certifications in cloud computing, applied machine learning, and modern web development, with **Git**-driven ownership from prototyping through **cloud deployment**.

Education

Vellore Institute of Technology

Bachelor of Technology, Computer Science & Engineering (AI & ML) | CGPA: **8.05/10.0**

Sept 2023 – May 2027

India

Montfort Inter College

Class 10 & 12 (Higher Secondary)

Apr 2021 – Mar 2023

Lucknow, India

Technical Skills

AI/ML: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Vector Databases (ChromaDB), LangChain, Prompt Engineering, Model APIs (Groq, Google Generative AI), AI Agent Design, Voice AI Agents, Scikit-learn
Languages: Python, Java

Backend & APIs: Node.js, REST API Development & Design, Authentication

Frontend: React.js, Next.js, HTML5, CSS3

Cloud & Databases: PostgreSQL, Supabase (DB & Auth), Database Design, Render Deployment

Tools & Practices: Git, GitHub, Dataset Development

Work Experience

Future Smart AI

Sept 2025 – Feb 2026

Content Research Intern | [Link](#)

Remote

- Evaluated and benchmarked more than **1,000 AI models and tools** across **4 categories** – video, image, captioning, and content generation – against marketing claims, producing a trusted, evidence-based capability assessment.
- Authored comparative technical reports scoring each model against **3 criteria** – strengths, limitations, and best-fit use case – helping teams select the right AI solution faster and with greater confidence.

Projects

Interview IQ | [Live Demo](#) | [GitHub](#)

Jul 2025 – Aug 2025

- Candidates lacked reliable prep – architected a document-grounded **RAG** pipeline with **LangChain** and a **vector database** for context-aware retrieval – delivered verified, source-backed answers.
- Prep felt generic – engineered a **PostgreSQL**-backed data layer logging history across **4 activity types** – tailored each candidate's study path with real signals.
- Prep resources were scattered – delivered **3 formats** of **LLM**-powered study material, **VAPI**-powered voice-AI interviews, and a resume gap analyzer – unified prep into one workflow.

AI-EchOo Grocery Platform | [Live Demo](#) | [GitHub](#)

Feb 2026

- Shoppers faced a static catalog – built a generative-AI shopping assistant across **6 purchase flows** – turned browsing into guided discovery and comparison.
- Assistant answers needed real-time context – designed a relational **4-table** schema and integrated **Groq**-hosted **LLMs** with **RAG** – anchored responses in live inventory.
- Shoppers needed hands-free ordering – implemented **Web Speech Recognition** for voice commands with real-time transcription – enabled natural, conversational shopping via automatic query submission.

Desktop AI Voice Assistant | [Live Demo](#) | [GitHub](#)

May 2025

- Manual input slowed tasks – developed a Python-based **voice AI agent** wiring **Groq** and **Google Generative AI** into **REST API** calls – enabled hands-free voice and text commands.
- Assistant felt robotic – designed natural **conversation flows** with two-way speech-to-text/text-to-speech via a **PyQt5** interface and **asynchronous processing** – automated browser workflows across **2 tasks**.
- Commands took multiple steps – optimized the inference pipeline to execute **4 categories** of commands – collapsed steps into one spoken instruction.

Achievements

Won 1st Place among **100+ participants** at the AI Demos Monthly Hackathon for an innovative AI solution. ([Link](#))

Published a Carbon Credits Dataset on Mendeley Data for a machine learning-based carbon credit prediction model.

([Dataset](#) | [Predictor](#))

Certifications

- Microsoft Azure Fundamentals (AZ-900)** – Cloud computing, cloud services, security fundamentals ([Certificate](#))
- Applied Machine Learning in Python** – University of Michigan, Coursera ([Certificate](#))
- MERN Stack** – MongoDB, Express.js, React.js, Node.js ([Certificate](#))